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Newton's second law states that the rate of change of momentum of a body is proportional to the force applied, and acts in the direction of the force. $F = ma$.

Speed is a scalar quantity (only magnitude), e.g. 40 km/hr.
Velocity is a vector quantity (magnitude and direction), e.g. 40 km/hr towards north.

The SI unit of electric current is the Ampere. One ampere is the current when one coulomb of charge flows through a conductor in one second.

Given $u = 0$, $a = 2 \text{ m/s}^2$, $t = 5\text{s}$.

$v = u + at = 10 \text{ m/s}$.

$s = ut + \frac{1}{2}at^2 = 25 \text{ m}$.

Ohm's law states $V = IR$ at constant temperature. $V = I \times R = 2 \times 10 = 20 \text{ Volts}$.

We know $v = u + at$. Average velocity = $(u+v)/2$. Distance $s =$ average velocity $\times$ time. Substituting: $s = ut + \frac{1}{2}at^2$.

The three states of matter are solid, liquid, and gas, differing in how tightly their particles are packed and how freely they move.

Mitosis produces two identical daughter cells for growth and repair. Meiosis produces four daughter cells with half the chromosome number, for reproduction.

Two ways to conserve water: rainwater harvesting, and promptly repairing leaking pipes and taps.

$$F = ma, v = u + at, s = ut + \frac{1}{2}at^2, V = IR, P = VI$$